Ahnaf Zareef

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Technical Skills

Languages: C, C++, Verilog, Python, Bash
Protocols: I2C, SPI, UART, I2S, CAN, USB, Ethernet
Tools: Keil, Quartus, Icarus Verilog, Yosys/nextpnr, Altium, ESP-IDF, OrCAD, MATLAB, Logic Analyzer
Platforms: ARM Cortex-M, ESP32, Tang Nano 9K FPGA, DE10-Lite, FreeRTOS, Linux

Experience

Firmware/ML Developer Intern — Feb 2026 – Present
Carobot Learning & Research Organization — *Toronto, ON*

- Developed a fully-offline robotics development-kit to build and deploy ML models on an **ESP32-S3** for vision and sensor based inference tasks on a block-based IDE.
- Developed **ESP32-S3 firmware** using **TinyUSB** on **ESP-IDF**, implementing a UVC + CDC composite device streaming OV2640 frames while receiving weights over serial.
- Restructured Blockly IDE with **TensorFlow.js**, implementing a block-based **NN** builder with configurable layers, activations, padding, and input dimensions for in-browser training.
- Implemented **Conv2D**, **Dense**, and **Pooling** forward-pass math from scratch in **Python**, generating MicroPython inference scripts for the ESP32-S3
- Evaluated ESP32 variants, ruling out non-S/P boards for insufficient **PSRAM** and lack of USB OTG required for UVC, and benchmarked MobileNet V1 at **2.1x** device RAM, pivoting to raw OV2640-trained models.

Power & Data Acquisition Member — Sep. 2025 – Present
McMaster Solar Car Project — *Hamilton, ON*

- Designed IV Curve Tracer **PCB** in **Altium** using a **Darlington pair** for **high-gain** current amplification, **ESP32** and a **Hall effect** sensor to characterize per-cell efficiency and detect faults.
- Developed protection circuitry: **MOSFET** switching on cell output, and **Zener** clamping on the **ESP32 ADC**.
- Developed ESP32 **C firmware** implementing a DAC-driven IV sweep with **16-sample ADC averaging**, settling delays for load and gate stabilization, and Hall effect **offset calibration**.

AI & ML Teaching Assistant — Jun 2025 – Sep 2025
Circuit Stream — *Toronto, ON*

- Taught supervised learning, CNNs, and MLPs to **40+** students using **scikit-learn**, **TensorFlow**, and OpenCV.
- Guided students building LLM chatbots with the **OpenAI API** and LangChain, debugging **RAG** grounding on custom data.

Projects

XNOR-9 | *FPGA, Verilog, Python, ESP32, C, RTL, UART, Yosys, nextpnr, Larq, TensorFlow*

- Built a **Binarized Neural Network accelerator** on a **Tang Nano 9K FPGA**, replacing multiply-accumulate with XNOR-popcount, achieving **32**× model compression.
- Hit **95% accuracy** within the 8,640-LUT budget by redesigning the parallel datapath for sequential neuron compute after weight pruning and layer removal fell short.
- Built an **ESP32-S3** front-end capturing hand-drawn digits, streaming them to the FPGA over **UART**.
- Reshaped weight storage to match the FPGA's **BRAM** geometry, cutting memory **47%** to fit 270K weights.

Lidar Mapper | *C, MATLAB, Keil, Logic Analyzer, MSP432, UART, I2C*

- Built **C** firmware for a spatial scanner, driving a half-stepped motor and VL53L1X ToF sensor over **I2C**, sampling 128 points per 360° sweep with per-step settling.
- Added interrupt-driven start/stop control and range validation, streaming polar data over **UART** to **MATLAB** for 3D point-cloud reconstruction.

Education

McMaster University — Hamilton, ON
Bachelor of Computer Engineering, (GPA: 3.5/4.0) — *Sep. 2024 – April 2028*